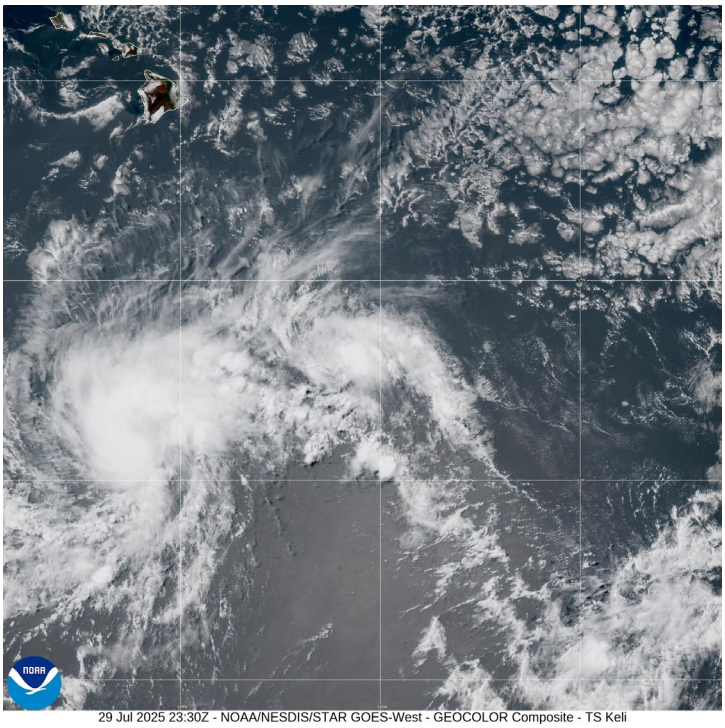
NATIONAL HURRICANE CENTER TROPICAL CYCLONE REPORT[1]

TROPICAL STORM KELI

(CP022025)

28–30 July 2025

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National Hurricane Center
16 October 2025



GOES-18 GEOCOLOR IMAGE OF TROPICAL STORM KELI (CENTER OF IMAGE) AT 2330 UTC 29 JULY. HURRICANE IONA IS LOCATED TO THE WEST-SOUTHWEST OF KELI. IMAGE COURTESY OF NOAA/NESDIS/STAR.

Keli was a small and short-lived tropical storm over the Central Pacific basin that co-existed with Hurricane Iona. It had no impact on any land area.

[1] This is an abbreviated Tropical Cyclone Report since there were no coastal watches or warnings issued and no direct fatalities reported in association with Keli.

Tropical Storm Keli

28–30 JULY 2025

BEST TRACK

The "best track[2]" positions and intensities for Tropical Storm Keli are listed in Table 1. The best track chart of Keli's path is given in Fig. 1, with the wind and pressure histories along with available observations[3] shown in Figs. 2 and 3, respectively.

There were no surface observations of winds of tropical storm force associated with Keli.

Origin

Keli formed from a low-pressure area that developed on 26 July during a breakdown of the Intertropical Convergence Zone about 1275 n mi east-southeast of Hilo, Hawaii. The low developed a small and well-defined circulation around 1800 UTC 27 July when it was located about 970 n mi east-southeast of Hilo, although the system did not have sufficient organized convection to be designated a tropical depression at that time. A subsequent increase in convective organization led to the formation of a tropical depression near 1200 UTC 28 July about 825 n mi southeast of Hilo.

Peak Intensity and Minimum Pressure

Keli's peak intensity of 45 kt from 0600–1200 UTC 29 July is based on a blend of scatterometer data and synthetic aperture radar (SAR) data (Fig. 4). While there remain concerns about SAR, especially the representativeness for estimating 1-minute peak sustained winds in tropical cyclones, a subsequent microwave pass (Fig. 4) also indicated the brief formation of inner

[2] A digital record of the complete best track, including wind radii, can be found on line at ftp://ftp.nhc.noaa.gov/atcf. Data for the current year's storms are located in the *btk* directory, while previous years' data are located in the *archive* directory.

[3] Observations include subjective satellite-based Dvorak technique intensity estimates from the Central Pacific Hurricane Center (PHFO), the Joint Typhoon Warning Center (JTWC) and the Satellite Analysis Branch (SAB), as well as objective Advanced Dvorak Technique (ADT) estimates and Satellite Consensus (SATCON) estimates from the Cooperative Institute for Meteorological Satellite Studies/University of Wisconsin-Madison. Data and imagery from NOAA polar-orbiting satellites including the Advanced Microwave Sounding Unit (AMSU), the NASA Global Precipitation Mission (GPM), the European Space Agency's Advanced Scatterometer (ASCAT), the Defense Meteorological Satellite Program (DMSP) satellites, and the Time-Resolved Observations of Precipitation structure and storm Intensity with a Constellation of Smallsats (TROPICS) satellites, among others, were also useful in constructing the best track of Keli.

core features that support a post-analysis increase of the peak intensity. The estimated minimum central pressure of 1003 mb is based on the Knaff-Zehr-Courtney pressure-wind relationship.

CASUALTY AND DAMAGE STATISTICS

There were no reports of damage or casualties associated with Keli.

FORECAST AND WARNING VERIFICATION

The genesis of Keli was essentially not forecast. Table 2 provides the number of hours in advance of formation with the first NHC Tropical Weather Outlook (TWO) forecast in each likelihood category. The system that became Keli was not mentioned in the TWO until a special TWO issued three hours before genesis occurred, at which time it was given a medium (40-60%) chance of development. The system was given a high (>60%) chance of development in a TWO issued at the time of genesis. While global models correctly showed that a low-pressure area would form to the east of developing Hurricane Iona, it was thought that this very small system would have little chance of development due to the presence of Iona to the west and another larger disturbance to the east.

Figure 5 shows composites of 7-day TWO genesis areas for each category prior to the formation of Keli. Note that while the genesis location falls within the forecast area, there was only one verifying forecast.

A verification of NHC official track forecasts for Keli is given in Table 3a. Official track forecast errors were significantly greater than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official track errors with selected guidance models is given in Table 3b. While the number of forecasts is small, the vast majority of the track guidance models had lower average errors than the official forecast. Examination of the individual forecasts (not shown) indicates that the initial track forecast showed a straight westward motion, and as Keli consistently moved north of that initial prediction, the official forecasts did not shift northward as quickly as the track guidance did.

A verification of NHC official intensity forecasts for Keli is given in Table 4a. Official intensity forecast errors were lower than the mean official errors for the previous 5-yr period. A homogeneous comparison of the official intensity errors with selected guidance models is given in Table 4b. While the official forecasts had lower average errors than the majority of the guidance, the number of forecasts is again small, and the majority of the average errors are less than the 5-kt precision of the NHC best track. The intensity forecasts correctly called for little intensification due to expected vertical shear and the proximity of Keli to Hurricane Iona.

There were no coastal watches or warnings issued for Keli.

ACKNOWLEDGEMENTS

The Central Pacific Hurricane Center in Honolulu, Hawaii, and Joint Typhoon Warning Center at Pearl Harbor, Hawaii, provided their fix information for Keli. Philippe Papin created the TWO verification graphic and the SAR/GPM data graphic.

Table 1. Best track for Tropical Storm Keli, 28–30 July 2025.

Date/Time (UTC)	Latitude (°N)	Longitude (°W)	Pressure (mb)	Wind Speed (kt)	Stage
27 / 1800	11.6	140.3	1010	25	low
28 / 0000	11.7	141.3	1010	25	"
28 / 0600	11.8	142.3	1009	30	"
28 / 1200	11.9	143.2	1008	30	tropical depression
28 / 1800	12.1	144.1	1006	35	tropical storm
29 / 0000	12.2	145.0	1004	40	"
29 / 0600	12.3	146.1	1003	45	"
29 / 1200	12.5	147.5	1003	45	"
29 / 1800	12.7	149.2	1004	40	"
30 / 0000	13.0	151.0	1006	35	"
30 / 0600	13.3	152.8	1007	35	"
30 / 1200	13.6	154.7	1007	35	"
30 / 1800					dissipated
28 / 1800	12.1	144.1	1006	35	minimum pressure

Table 2. Number of hours in advance of formation associated with the first NHC Tropical Weather Outlook forecast in the indicated likelihood category. Note that the timings for the “Low” category do not include forecasts of a 0% chance of genesis.

	Hours Before Genesis	
	48-Hour Outlook	**168-Hour Outlook**
Low (<40%)	N/A	N/A
Medium (40%-60%)	3	3
High (>60%)	0	0

Table 3a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) track forecast errors (n mi) for Tropical Storm Keli, 28–30 July 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	27.3	70.4	120.1	205.2				
OCD5	54.0	118.9	162.4	158.6				
Forecasts	7	5	3	1	0	0	0	0
OFCL (2020-24)	22.4	33.6	43.5	53.2	65.1	79.4	108.6	135.2
OCD5 (2020-24)	38.2	75.3	117.2	161.0	206.0	252.0	340.7	422.7

Table 3b. Homogeneous comparison of selected track forecast guidance models (in n mi) for Tropical Storm Keli, 28–30 July 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 3a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	29.0	68.7	111.6					
OCD5	58.9	128.8	205.5					
GFSI	**27.8**	**54.0**	**73.6**					
HWFI	35.0	76.2	**101.4**					
HMNI	**25.9**	**63.4**	**89.4**					
HFAI	**16.5**	**34.0**	**48.5**					
HFBI	**19.7**	**39.2**	**60.8**					
EMXI	**16.5**	**19.7**	**15.8**					
CTCI	36.0	92.4	157.6					
TVCE	**20.5**	**50.7**	**72.3**					
TVCX	**19.7**	**42.6**	**65.8**					
GFEX	**20.7**	**25.9**	**38.1**					
TVDG	**20.2**	**44.0**	**64.8**					
HCCA	**16.5**	**32.5**	**64.9**					
FSSE	**16.0**	**29.0**	**42.9**					
AEMI	**24.1**	**52.1**	**85.4**					
TABS	**25.1**	**48.6**	**86.4**					
TABM	**28.8**	**47.0**	**57.0**					
TABD	31.7	**49.6**	**64.8**					
Forecasts	6	3	2	0	0	0	0	0

Table 4a. NHC official (OFCL) and climatology-persistence skill baseline (OCD5) intensity forecast errors (kt) for Tropical Storm Keli, 28–30 July 2025. Mean errors for the previous eastern Pacific 5-yr period are shown for comparison. Official errors that are smaller than the 5-yr means are shown in boldface type.

	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	**2.1**	**1.0**	**1.7**	**0.0**				
OCD5	3.1	5.8	11.7	9.0				
Forecasts	7	5	3	1				
OFCL (2020-24)	5.7	8.9	10.8	12.9	14.4	15.5	17.0	18.6
OCD5 (2020-24)	7.4	12.4	16.2	19.2	21.0	21.3	21.5	21.6

Table 4b. Homogeneous comparison of selected intensity forecast guidance models (in kt) for Tropical Storm Keli, 28–30 July 2025. Errors smaller than the NHC official forecast are shown in boldface type. The number of official forecasts shown here will generally be smaller than that shown in Table 4a due to the homogeneity requirement.

Model ID	Forecast Period (h)							
	12	24	36	48	60	72	96	120
OFCL	2.5	1.7	0.0					
OCD5	3.0	6.3	13.0					
HWFI	4.5	2.3	1.0					
HMNI	5.0	2.0	1.0					
HFAI	5.2	2.3	0.5					
HFBI	5.5	3.0	2.0					
DSHP	3.2	2.3	7.0					
LGEM	3.5	2.3	6.0					
ICON	3.7	**1.3**	4.0					
IVCN	4.2	1.7	2.5					
IVDR	4.3	2.0	2.0					
CTCI	4.3	2.7	2.0					
GFSI	6.5	4.0	1.0					
EMXI	5.0	2.3	0.0					
HCCA	3.7	5.0	8.5					
FSSE	3.5	**1.0**	2.5					
Forecasts	6	3	2	0	0	0	0	0

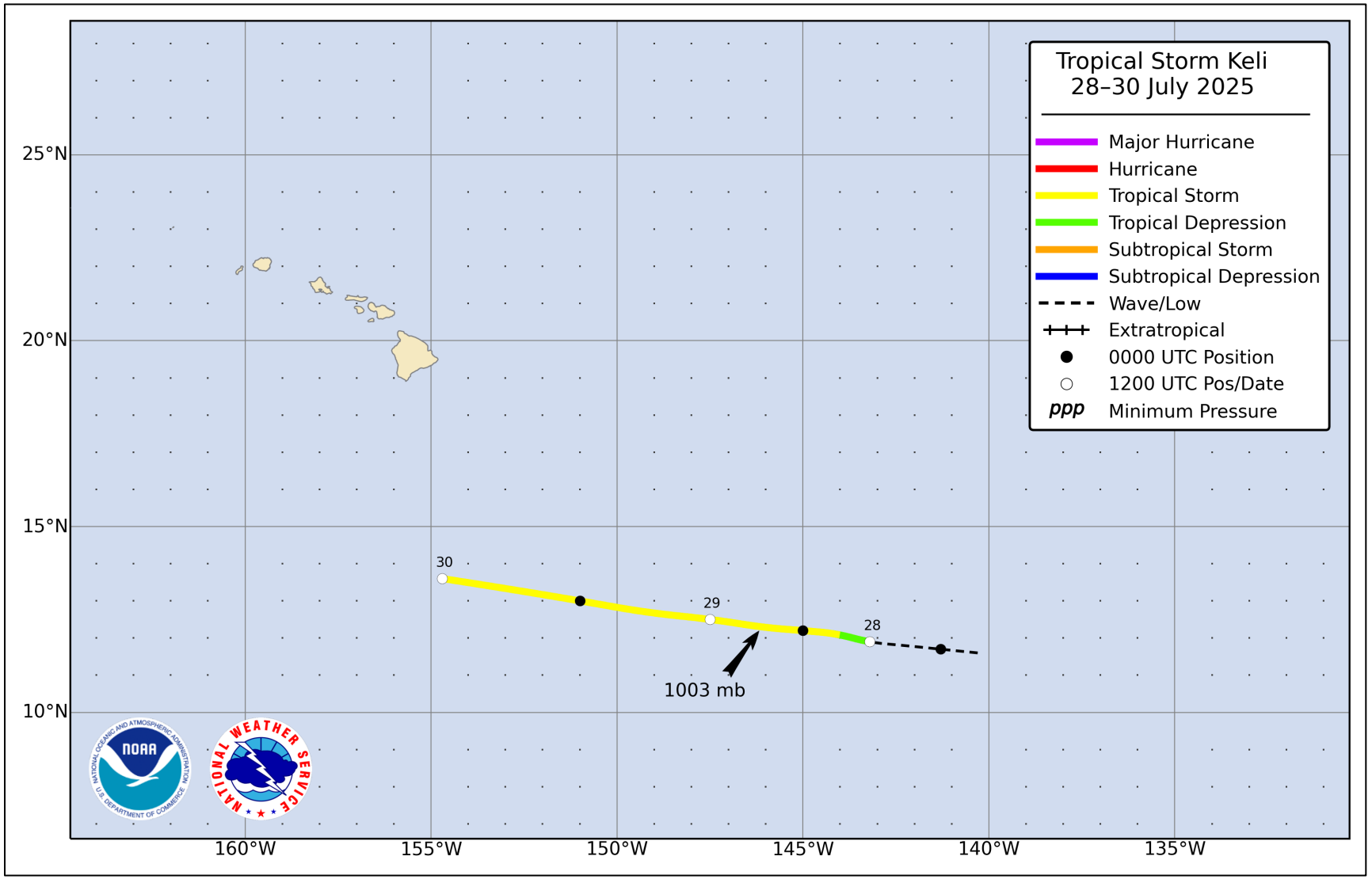


Figure 1. Best track positions for Tropical Storm Keli, 28–30 July 2025.

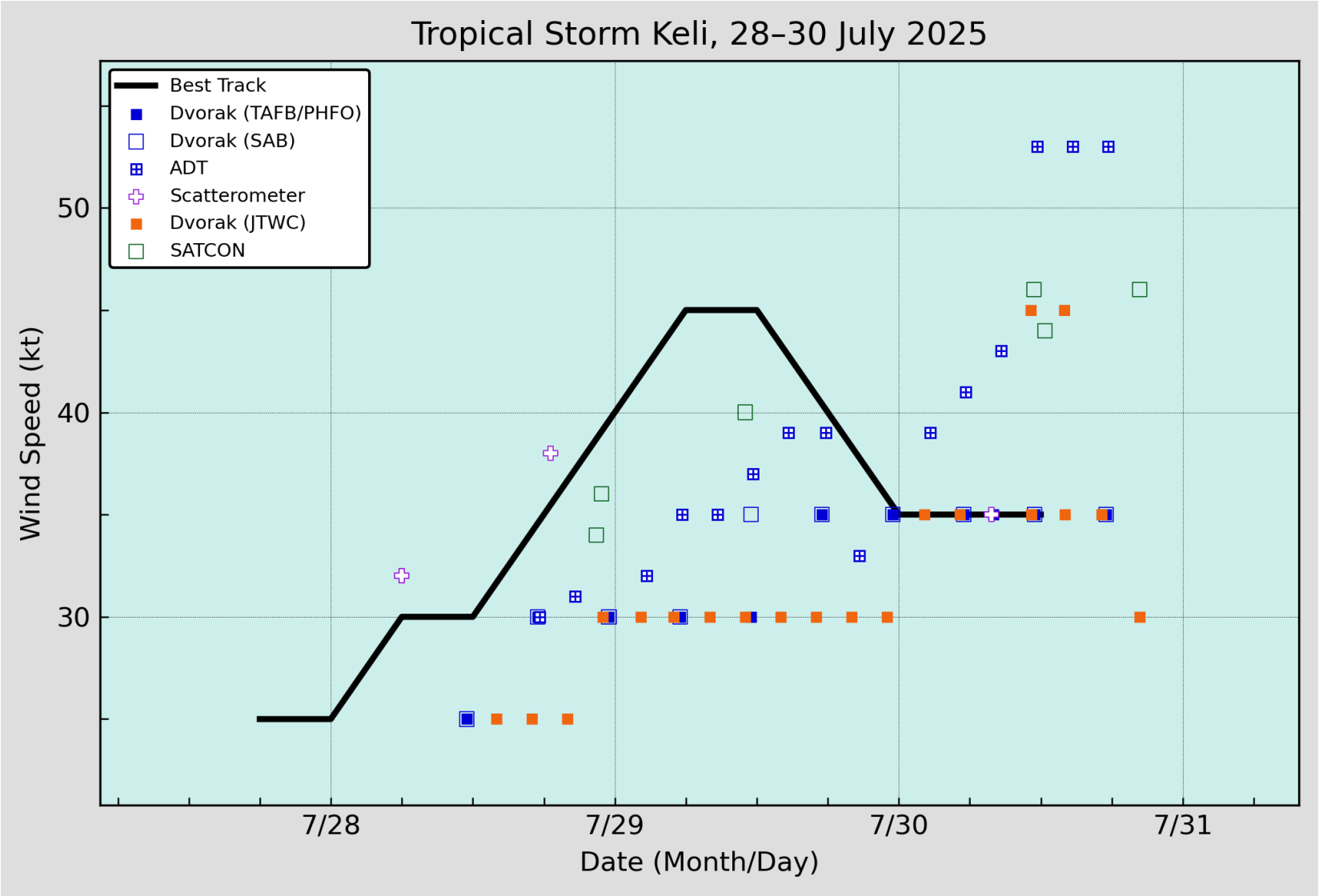


Figure 2. Selected wind observations and best track maximum sustained surface wind speed curve for Tropical Storm Keli, 28–30 July 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. Dashed vertical lines correspond to 0000 UTC.

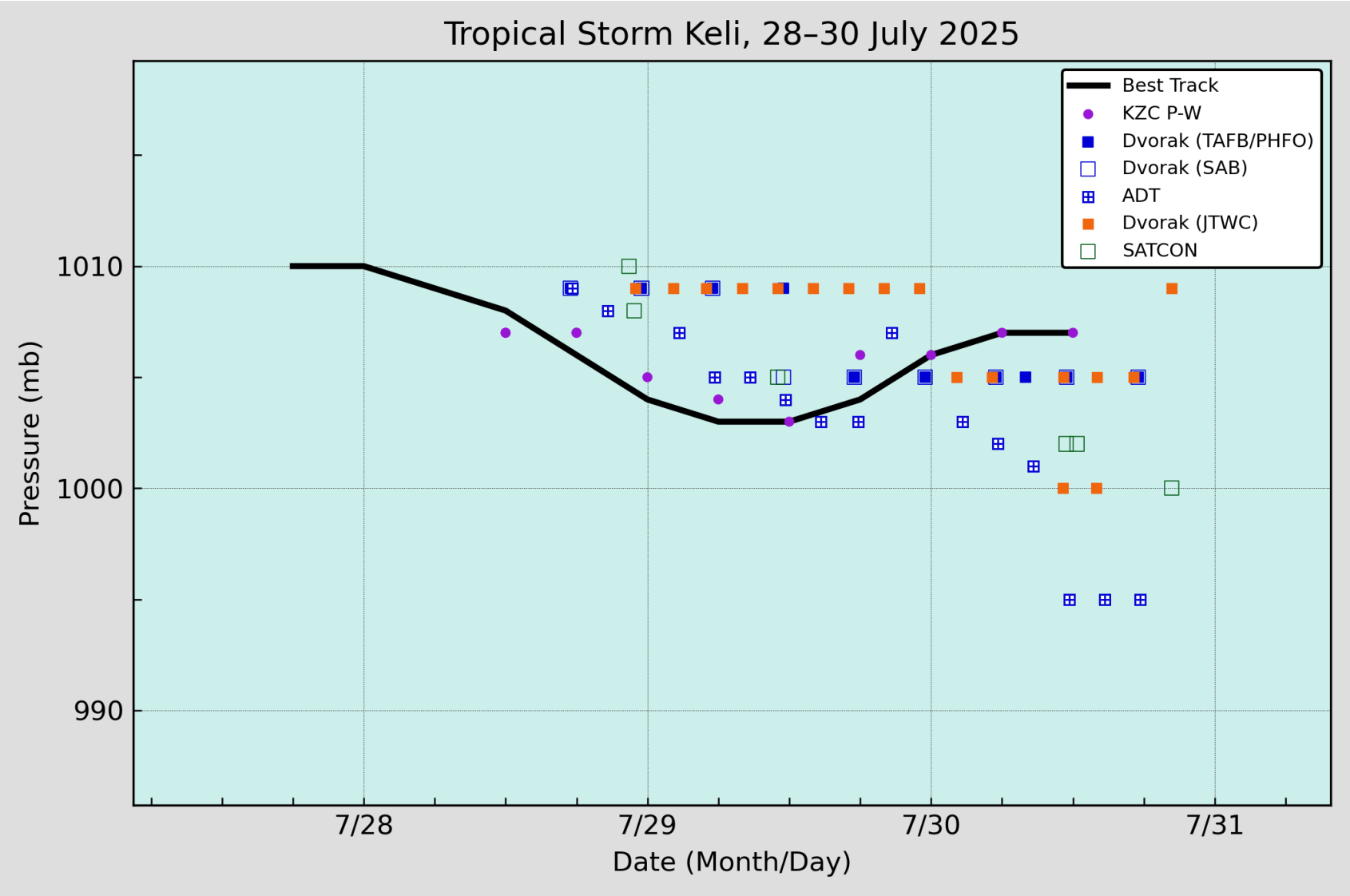


Figure 3. Selected pressure observations and best track minimum central pressure curve for Tropical Storm Keli, 28–30 July 2025. Advanced Dvorak Technique estimates represent the Current Intensity at the nominal observation time. SATCON intensity estimates are from the Cooperative Institute for Meteorological Satellite Studies. KZC P-W refers to pressure estimates derived using the Knaff-Zehr-Courtney pressure-wind relationship. Dashed vertical lines correspond to 0000 UTC.

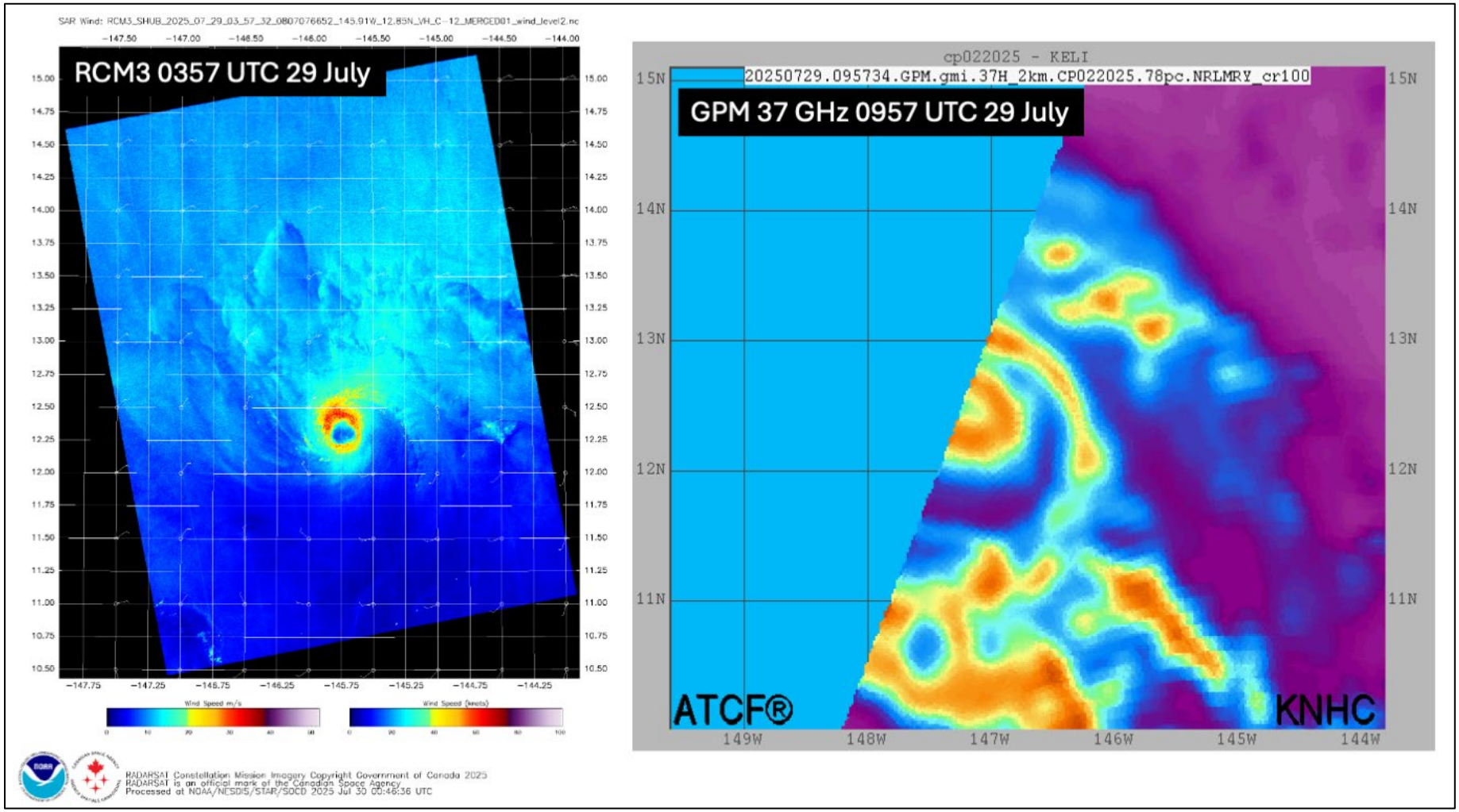


Figure 4. (Left) Synthetic aperture radar (SAR) wind speed data for Keli from the RCM-3 satellite at 0357 UTC 29 July courtesy of NESDIS STAR, and (Right) 37 GHz brightness temperature data for Keli from the GPM satellite at 0957 UTC 29 July courtesy of NRL Monterey. Note that the calibration of wind estimates from SAR are still being examined, and the data were only used qualitatively when estimating Keli's peak intensity.

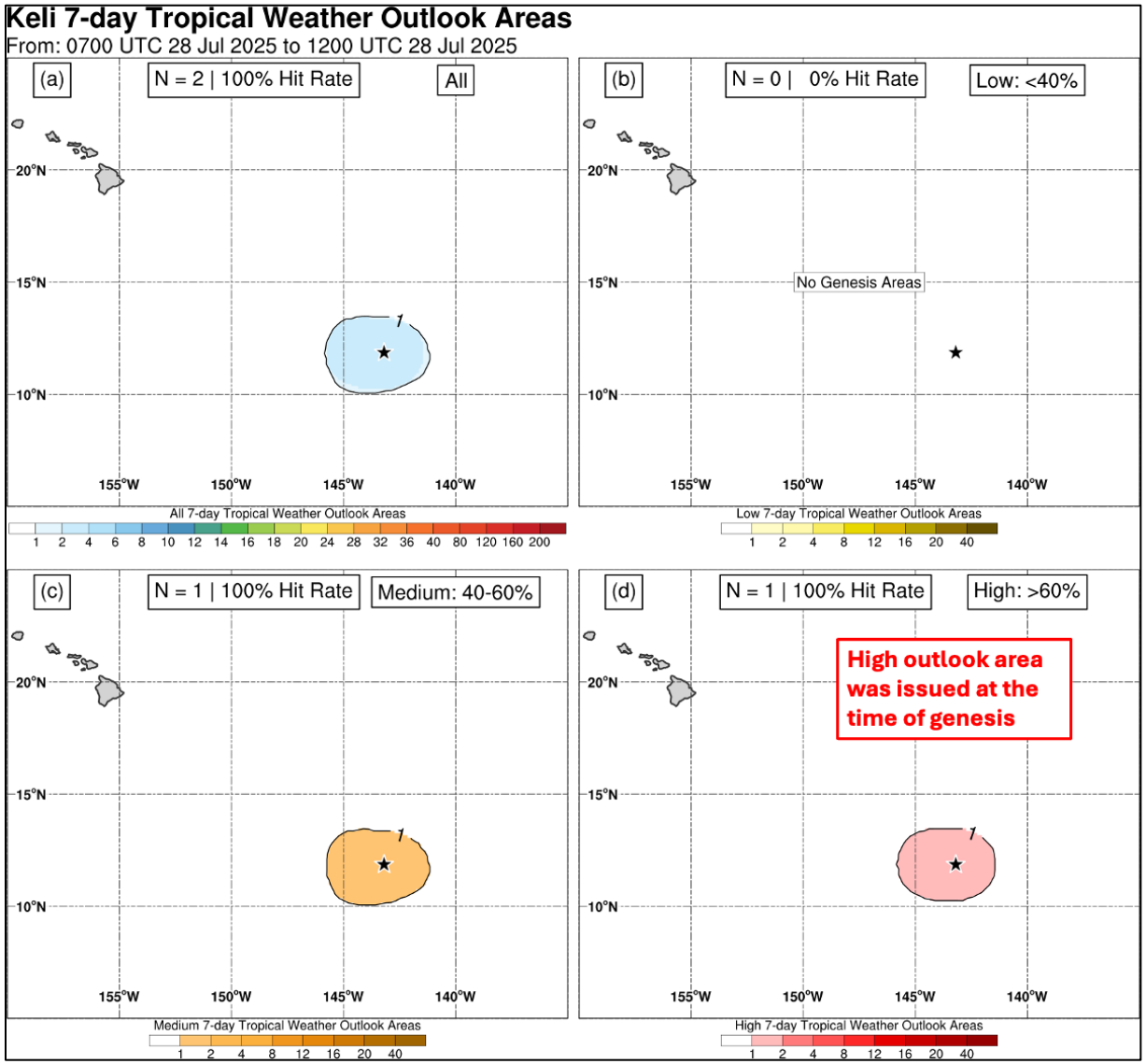


Figure 5. Composites of 7-day tropical cyclone genesis areas depicted in NHC's Tropical Weather Outlooks prior to the formation of Tropical Storm Keli, 28–30 July 2025 for (a) all probabilistic genesis categories, (b) the low (<40%) category, (c) medium (40–60%) category, and (d) high (>60%) category. The location of genesis is indicated by the black star.